

Sensitivity study for $B^0 \rightarrow K^{*0}(K^+\pi^-)\ell^+\ell^-$ (v2)

Belle II-like Sensitivity Study Framework
Analysis agent, AI-assisted write-up

Version 2 – September 19, 2026

Public materials disclaimer. This note is the product of an automated sensitivity-study framework using fast simulation and simplified detector modeling. It is *not* a Belle II collaboration result and must not be represented as such. All numbers derive from Monte Carlo generated within this framework’s tools during this and a prior session, as documented herein.

Abstract

We evaluate the counting and template-fit sensitivity of a Belle II-like detector to $B^0 \rightarrow K^{*0}(K^+\pi^-)\ell^+\ell^-$ ($\ell = e, \mu$) at an integrated luminosity of 1 ab^{-1} , following the reconstruction and background-suppression strategy of Wei et al. [Belle] [1] and Belle II [2]. Using the frozen selection produced in a prior optimization pass of this framework (K^{*0} mass window, Fox–Wolfram R_2 continuum suppression, flavor-dependent ΔE windows, and asymmetric $J/\psi/\psi(2S)$ vetoes in $m(\ell^+\ell^-)$), the signal efficiency is 34.8% (e) and 31.6% (μ) relative to generated events. In the beam-constrained-mass counting signal region ($M_{\text{bc}} > 5.27 \text{ GeV}$, on top of all other cuts) the expected weighted yield is $S = 491.7$ signal events against a background dominated by generic- $B\bar{B}$ combinatorics, $B = 1650$ events from only 3 raw simulated MC candidates (68% CL Poisson interval [752, 3255], i.e. $+1605/-898$), giving a nominal $S/\sqrt{S+B} = 10.6$ that ranges from 8.0 to 13.9 across the Poisson interval on B alone. A binned maximum-likelihood fit to the M_{bc} shape over the full sideband-inclusive range [5.20, 5.29] GeV, split by lepton flavor and sharing a common signal strength μ , recovers $\mu = 1.000 \pm 0.057$ at 1 ab^{-1} ($\mu = 1.000 \pm 0.062$ electron-only, 1.000 ± 0.145 muon-only), i.e. a 5.7% relative precision on the branching fraction at this luminosity. This Asimov-style check validates only that the fit is unbiased and returns $\mu = 1$ by construction; no toy/pull study was run in this session. A statistics-only Asimov projection gives 17.4% relative precision at 0.1 ab^{-1} , 9.2% at 0.36 ab^{-1} , 5.5% at 1 ab^{-1} , 2.5% at 5 ab^{-1} and 0.8% at 50 ab^{-1} . The $J/\psi K^{*0}$ control channel is a very high-purity control sample (>99% using the dedicated J/ψ MC and a representative combinatorial file; >90% under a conservative scaling assumption). The main limitation is the size of the combinatorial background MC samples.

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1 Introduction

$B^0 \rightarrow K^{*0} \ell^+ \ell^-$ is a flavor-changing-neutral-current $b \rightarrow s \ell^+ \ell^-$ decay, loop-suppressed and highly sensitive to new physics contributions that can alter its branching fraction, angular distributions and lepton universality ratio relative to the Standard Model. We follow the reconstruction and background-control strategy of the first statistically significant Belle measurement of this mode, Wei et al. [Belle], PRL 103, 171801 (2009) [1]: a K^{*0} mass window around 0.896 GeV, a beam-constrained mass $M_{bc} > 5.27$ GeV, and an energy-difference ΔE window that is asymmetric and wider on the low side for electrons because of bremsstrahlung; charmonium backgrounds $B \rightarrow J/\psi(\psi(2S))K^{*0}$ are removed with asymmetric vetoes in the dilepton mass (again wider on the low side for e^+e^-), and the vetoed $J/\psi K^{*0}$ sample serves as the control channel for efficiencies and particle identification. The Belle II update at 189 fb⁻¹ (arXiv:2206.05946) [2] uses the same overall strategy with explicit veto windows – J/ψ excluded for $m(\mu^+\mu^-) \in [2.946, 3.176]$ GeV and $m(e^+e^-) \in [2.846, 3.176]$ GeV, with the $\psi(2S)$ vetoed similarly at its own mass – and extracts the signal from a 2D unbinned M_{bc} - ΔE fit with an ARGUS-plus-linear combinatorial model. This study adopts the Belle II veto windows exactly and follows both papers’ qualitative pattern of asymmetric, electron-wider ΔE and charmonium-veto windows (confirmed numerically in Section 3). Unlike the published analyses we perform a 1D M_{bc} fit only (no ΔE dimension) and model all combinatorial background categories (generic $B\bar{B}$ and continuum) with dedicated MC samples rather than fitting an ARGUS parameterization to data; both simplifications are discussed in Section 7.

This note is a write-up pass over an analysis whose selection scans, cutflow optimization and figure production were completed in a prior session of this framework; the present session re-verifies the key numbers with a small number of counting queries and the two frozen template fits, and assembles the final note.

2 Data samples and analysis tools

All samples are generated with the framework’s EvtGen + fast-simulation chain (signal, $J/\psi K^{*}$ background) or with Pythia8 continuum generation piped directly to ntuples (continuum combinatorial), and processed into flat ROOT ntuples with `make_ntuple_exclusive.py` (signal and dedicated J/ψ background) or `make_ntuple_combinatorial.py` / `generate_continuum_exclusive.py` (combinatorial candidates). Individual random seeds used when these files were produced in the prior session were not communicated to this write-up pass; since no new MC is generated here (per the scope of this pass) they cannot be recovered without re-running the generation, which is explicitly out of scope. This is recorded as a reproducibility gap in Section 7.

Combinatorial candidate definition: every $K^+\pi^-\ell^+\ell^-$ combination in a generic $B\bar{B}$ or continuum event that falls in the loose windows $|m(K\pi) - 0.896| < 0.25$ GeV, $M_{bc} > 5.20$ GeV, $|\Delta E| < 0.40$ GeV is kept as a candidate; kaon/pion species are assigned from MC truth (no K/π mis-identification is modeled) and lepton fakes are included via the same fast-simulation lepton

identification used for signal. Each candidate additionally carries `11_true_src`, which is 1 if both leptons genuinely come from the same true J/ψ or $\psi(2S)$ decay and 0 otherwise. Whenever a combinatorial sample is combined with the dedicated $J/\psi K^{*0}$ MC sample in a composition or control-region table, `11_true_src==0` is required on the combinatorial file to avoid double-counting; this requirement is *not* applied in the global M_{bc} template fit, where the $m(\ell\ell)$ veto already removes essentially all resonant charmonium from the combinatorial templates (Section 6).

Table 1: Samples used in this study, their generated statistics, and the weight applied to scale each file to 1 ab^{-1} .

File(s)	mode_id	Generated events	Weight to 1 ab^{-1}
sig_kstll.root	20	10,000 (5,000 e + 5,000 μ)	0.1468 (ee) / 0.1497 ($\mu\mu$)
bkg_jpsikst.root	21	5,000	21.618
comb_generic_0..7.root (8 files)	97	2,000,000 each	550 / file
comb_continuum_0..1.root (2 files)	98	3,000,000 each	876.3 / file

Weight sources: signal, $N(B^0\bar{B}^0) = 2 \times 0.486 \times 1.1 \text{ nb} \times 10^9 \text{ nb}^{-1}$, $\text{BF}(K^{*0}e^+e^-) = 1.03 \times 10^{-6}$, $\text{BF}(K^{*0}\mu^+\mu^-) = 1.05 \times 10^{-6}$, $\text{BF}(K^{*0} \rightarrow K^+\pi^-) = 2/3$; $J/\psi K^{*0}$, $\text{BF}(J/\psi K^{*0}) = 1.27 \times 10^{-3} \times \text{BF}(J/\psi \rightarrow \ell\ell) = 0.0597$; generic $B\bar{B}$, $1.1 \times 10^9/2 \times 10^6$; continuum, $\sigma_{q\bar{q}} = 2.629 \text{ nb}$: $2.629 \times 10^9/3 \times 10^6$.

Total simulated combinatorial statistics: 16,000,000 generic $B\bar{B}$ events + 6,000,000 continuum events. As shown below, only a handful of these events survive the full signal selection, which is the dominant limitation of this study.

3 Event selection

The selection below is **frozen** from the prior optimization pass of this framework (scans are not repeated here, only single-point marked distributions are shown); each cut is quoted with the background it targets and the figure that motivated it.

1. **K^{*0} mass window**, $0.796 < m(K\pi) < 0.996 \text{ GeV}$ (`m_visible`): selects the $K^{*0}(892)$ resonance and rejects non-resonant $K\pi$ combinatorics and wrong-species pairings (Fig. 1).
2. **Continuum suppression**, $R_2 < 0.5$ (Fox–Wolfram ratio): rejects $e^+e^- \rightarrow q\bar{q}$ continuum, whose jet-like topology gives large R_2 values compared to the more spherical $B\bar{B}$ topology (Fig. 2).
3. **M_{bc} shape after the K^{*0} window** (diagnostic, motivates the $M_{bc} > 5.27 \text{ GeV}$ counting-region cut and the fit range used later, Fig. 3).
4. **Energy difference ΔE** , flavor-dependent and asymmetric: $-0.10 < \Delta E < 0.06 \text{ GeV}$ (electrons), $-0.08 < \Delta E < 0.06 \text{ GeV}$ (muons). This cut targets combinatorial background from generic $B\bar{B}$ and continuum, whose random $K\text{--}\pi\text{--}\ell\text{--}\ell$ combinations populate ΔE broadly, while correctly reconstructed signal peaks near $\Delta E = 0$; the wider low-side window for electrons accommodates final-state bremsstrahlung. Both windows match the asymmetric, electron-wider pattern of Wei et al. and Belle II (Figs. 4, 5).
5. **Charmonium vetoes** in $m(\ell\ell)$, asymmetric and wider for electrons: veto $m(ee) \in [2.846, 3.176] \text{ GeV}$ (J/ψ) and $[3.486, 3.766] \text{ GeV}$ ($\psi(2S)$); veto $m(\mu\mu) \in [2.946, 3.176] \text{ GeV}$ (J/ψ) and $[3.586, 3.766] \text{ GeV}$

($\psi(2S)$). This targets $B \rightarrow J/\psi(\psi(2S))K^{*0}$, whose rate is roughly $1000\times$ the signal rate. The low-side veto width relative to the J/ψ mass (3.0969 GeV) is 0.251 GeV for ee vs. 0.151 GeV for $\mu\mu$, wider for electrons as in the cited papers, while the high-side width (0.079 GeV) is common to both flavors (Figs. 6, 7).

6. **Counting signal region:** in addition to all the above, $M_{bc} > 5.27$ GeV, targeting combinatorial background under the M_{bc} peak (Fig. 8).

Anti-fluctuation rule: during the original optimization, a candidate working point was only accepted if it retained at least 10 raw (unweighted) background MC events, to avoid tuning cuts against single-MC-event statistical fluctuations. As shown in Section 5, the *final* combined signal region nonetheless ends up with only 3 raw weighted-550 combinatorial events (comb_generic) and 0 raw weighted-876.3 events (comb_continuum): the individual cuts were each validated against ≥ 10 raw events at the stage they were introduced, but their cumulative effect, once all 10 background files are summed, leaves the final region governed by single-digit raw statistics.

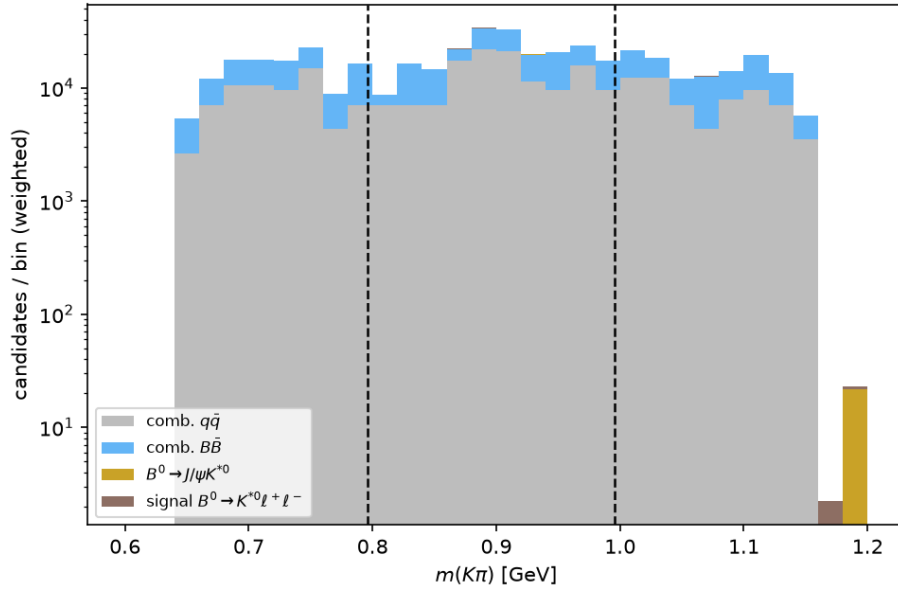


Figure 1: $m(K\pi)$ distribution at preselection, stacked by sample, showing the $K^{*0}(892)$ peak used to define the 0.796–0.996 GeV window (dashed lines) against the combinatorial and $J/\psi K^*$ components.

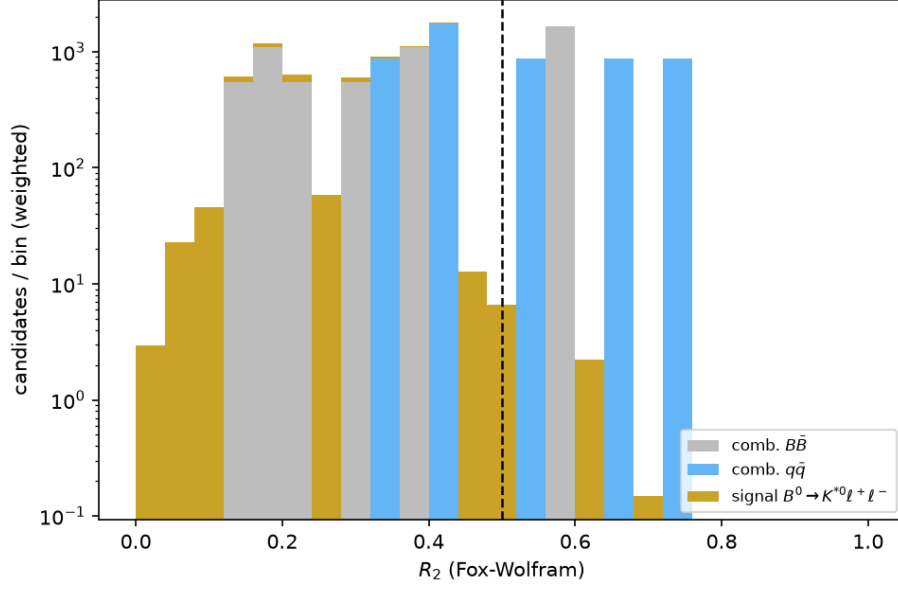


Figure 2: Fox-Wolfram R_2 distribution after the K^{*0} window, stacked by sample, showing the continuum component peaking at high R_2 and the $R_2 < 0.5$ working point (dashed line).

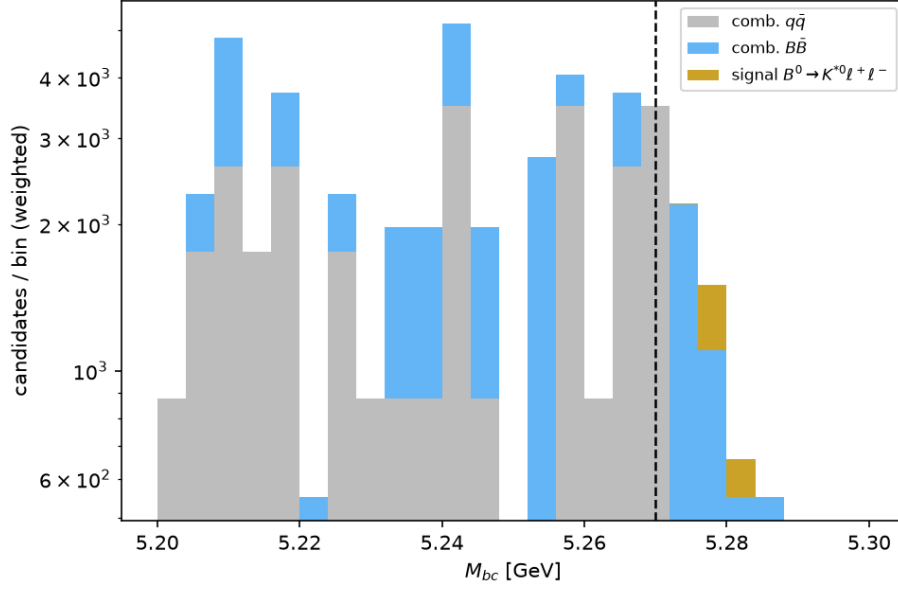


Figure 3: M_{bc} distribution after the K^{*0} window and R_2 cut, stacked by sample, showing the narrow signal peak near the B^0 mass on top of the combinatorial continuum that motivates the M_{bc} fit / counting-region choice.

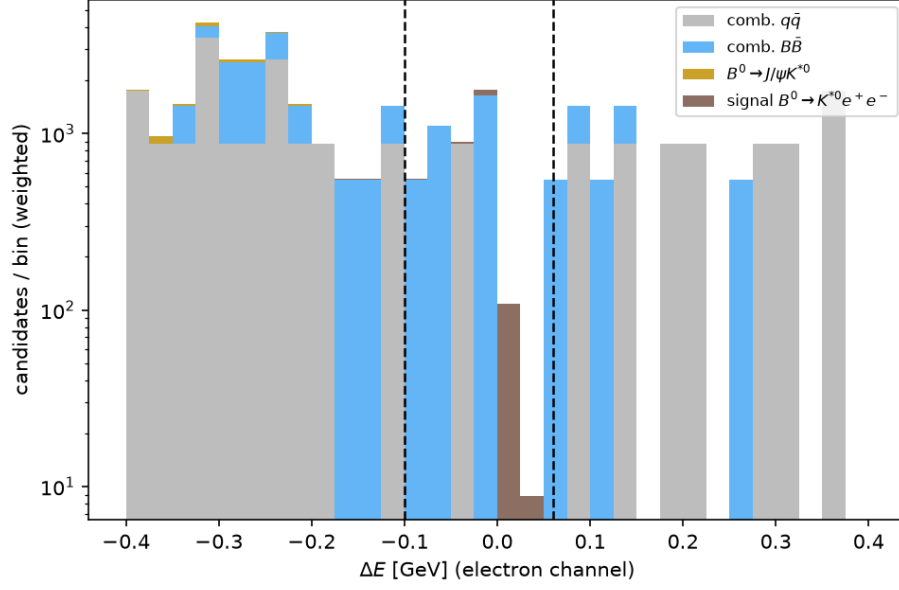


Figure 4: ΔE distribution for the electron channel, stacked by sample, with the $-0.10/+0.06$ GeV working point indicated.

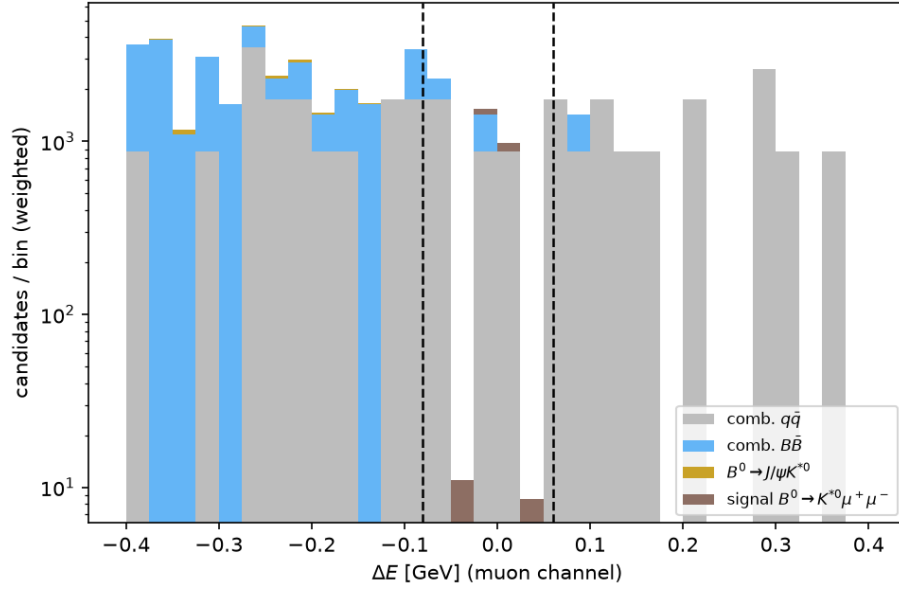


Figure 5: ΔE distribution for the muon channel, stacked by sample, with the $-0.08/+0.06$ GeV working point indicated.

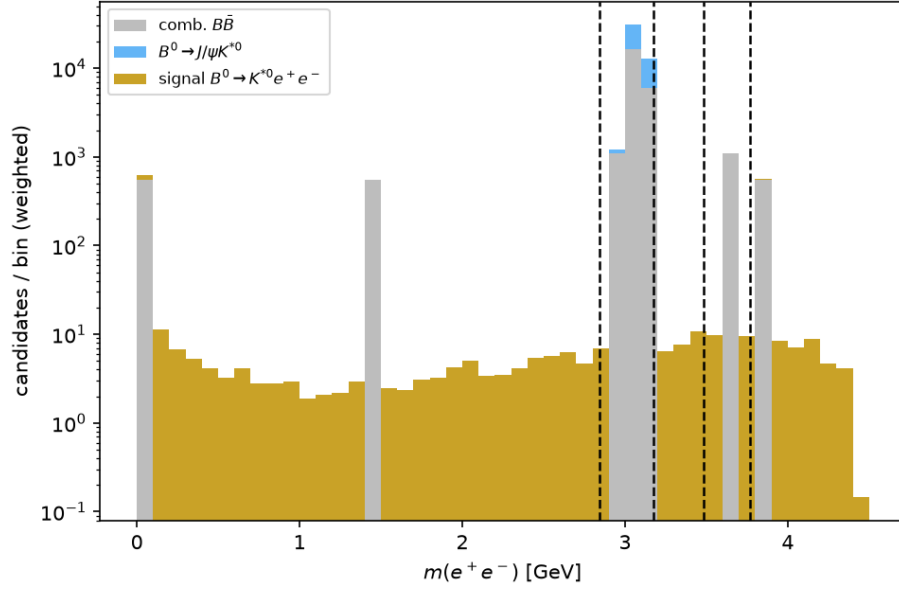


Figure 6: Dilepton mass for the electron channel, stacked by sample, with the J/ψ and $\psi(2S)$ veto windows indicated by dashed lines.

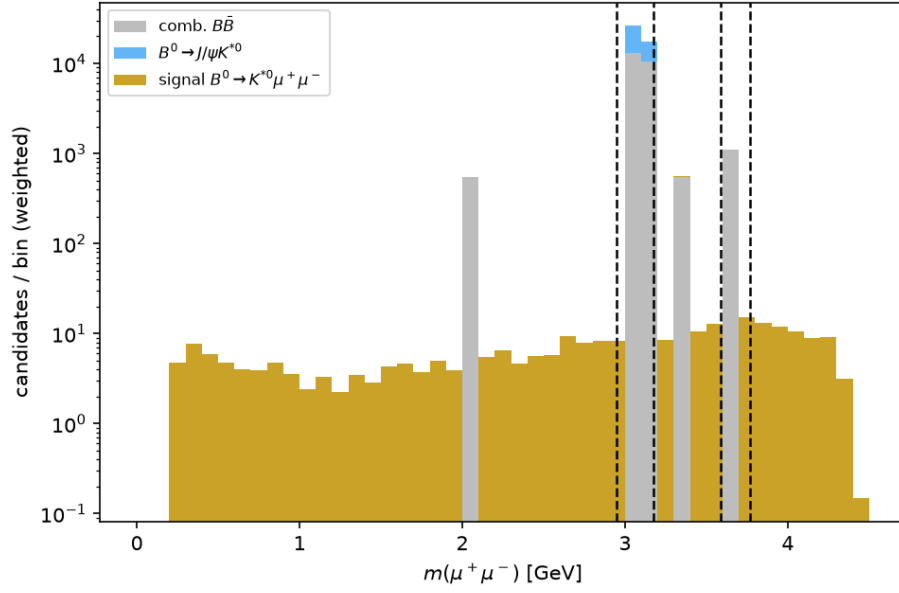


Figure 7: Dilepton mass for the muon channel, stacked by sample, with the J/ψ and $\psi(2S)$ veto windows indicated by dashed lines.

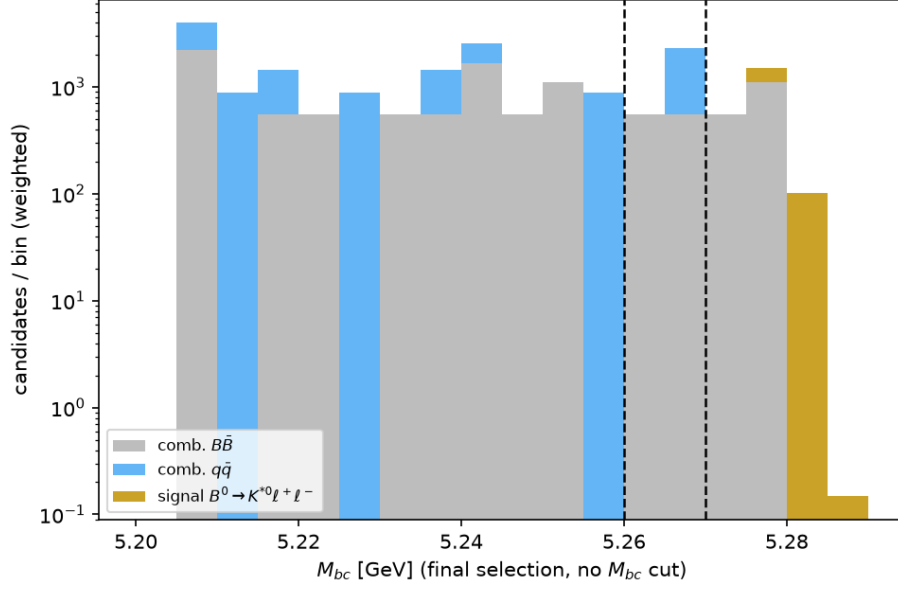


Figure 8: M_{bc} distribution at the final selection (all cuts applied except the M_{bc} cut itself), stacked by sample, showing the surviving signal peak and the combinatorial background used both for the $M_{bc} > 5.27$ GeV counting region and as the fit range.

4 Cutflow

This section quantifies the efficiency lost at each selection stage, so that the impact of each cut introduced above can be judged quantitatively rather than only qualitatively from the stacked distributions.

Table 2: Signal cutflow (sig_kstll.root, both flavors combined; efficiencies relative to 10,000 generated events).

Stage	N (weighted-1 events)	Cumulative eff.	Relative eff.
Generated	10,000	100.00%	–
Candidate reconstructed	4,993	49.93%	49.93%
+ K^{*0} window	4,386	43.86%	87.84%
+ $R_2 < 0.5$	4,276	42.76%	97.49%
+ ΔE window	3,930	39.30%	91.91%
+ charmonium veto	3,318	33.18%	84.43%
+ $M_{bc} > 5.27$ (counting SR)	3,318	33.18%	100.00%

The $M_{bc} > 5.27$ GeV cut removes essentially no additional signal, since correctly reconstructed $B^0 \rightarrow K^{*0} \ell^+ \ell^-$ candidates already peak at the B^0 mass with a narrow detector resolution.

Table 3: Per-flavor efficiency in the counting signal region (of 5,000 generated events per flavor).

Flavor	N passing SR	Efficiency
e^+e^-	1,740	34.80%
$\mu^+\mu^-$	1,578	31.56%

5 Background estimation

Three background categories are considered: the dedicated $B^0 \rightarrow J/\psi K^{*0}(\rightarrow \ell\ell)$ MC sample (a resonant peaking background removed almost entirely by the charmonium veto and used as the control channel), generic- $B\bar{B}$ combinatorial $K\pi\ell\ell$ candidates, and continuum- $q\bar{q}$ combinatorial $K\pi\ell\ell$ candidates (`11_true_src==0` required on both combinatorial categories to avoid double-counting genuine charmonium decays already modeled by the dedicated J/ψ sample).

Because the generic- $B\bar{B}$ component is estimated from only 3 raw MC events, its uncertainty is quoted as an exact 68% CL Poisson (Garwood) interval on the raw count, propagated through the per-event weight of 550, rather than a symmetric Gaussian $\sqrt{\sum w^2}$, which would understate the true asymmetric spread at such low counts (the Gaussian value is given alongside for reference).

Table 4: Weighted composition of the counting signal region (all cuts including $M_{bc} > 5.27$ GeV; raw counts summed over all files in each category).

Component	Raw N	Weighted N	68% CL interval	$\sqrt{\sum w^2}$ (ref.)
Signal ($K^{*0}\ell\ell$)	3,318	491.7	—	—
$J/\psi K^{*0}(\ell\ell)$	0	0.0	95% CL UL ~ 64.9	—
Generic $B\bar{B}$ combinatorial	3	1650.0	[752.0, 3255.0]	± 952.6
Continuum combinatorial	0	0.0	95% CL UL ~ 2628.9	—
Total background	3	1650.0	[752.0, 3255.0]	± 952.6

The J/ψ veto is fully effective within the available MC statistics: 0 of 2,396 reconstructed $J/\psi K^{*0}$ candidates survive the full selection including the veto, versus 1,886 surviving in the control region defined below (before the veto), so no continuous- M_{bc} J/ψ leakage is observed. The generic- $B\bar{B}$ combinatorial background dominates but is known only from 3 raw simulated events out of 16×10^6 generated $B\bar{B}$ events; the continuum contribution is compatible with zero given 0 raw events out of 6×10^6 generated events.

Table 5: $J/\psi K^{*0}$ control region (veto replaced by requiring $m(\ell\ell)$ inside the J/ψ window; $M_{bc} > 5.27$ GeV kept).

Component	Raw N	Weighted N
$J/\psi K^{*0}$ (dedicated MC)	1,886	40,771.5
Generic $B\bar{B}$ (comb_generic_0.root, representative)	1	550.0
Continuum (comb_continuum_0.root, representative)	0	0.0

Using the representative single file per combinatorial category, the control-region purity is $40,771.5/(40,771.5 + 550.0 + 0.0) = 98.7\%$. Conservatively scaling the representative combinato-

rial yield by the number of files in each category (8 and 2), the purity is $40,771.5/(40,771.5 + 4,400.0 + 0.0) = 90.3\%$. Either way this is a high-purity, high-statistics control sample suitable for calibrating lepton and K^{*0} reconstruction efficiencies in a real analysis, in sharp contrast to the MC-statistics-limited signal region itself.

M_{bc} **sideband** ($5.20 < M_{bc} < 5.26$ GeV, all other cuts applied, representative files only): comb_generic_0.root gives 2 raw / 1,100.0 weighted events, comb_continuum_0.root gives 3 raw / 2,628.9 weighted events. This sideband is the region used (together with the rest of the $[5.20, 5.27]$ GeV range) to constrain the combinatorial M_{bc} shape in the template fit below, so that the fit is not performed inside a region whose shape has itself been sculpted by the $M_{bc} > 5.27$ GeV cut.

6 Results

This section reports both a simple cut-and-count estimate in the M_{bc} signal region and a binned M_{bc} template fit that uses the full sideband range; the fit is taken as the primary result because it is far less sensitive to the low combinatorial MC statistics discussed above.

Cut-and-count: with $S = 491.7$ and $B = 1650.0$ (68% CL Poisson interval $[752.0, 3255.0]$) in the $M_{bc} > 5.27$ GeV signal region, the nominal significance estimator is $S/\sqrt{S+B} = 10.6$, ranging from 8.0 (B at the upper edge of its interval) to 13.9 (B at the lower edge). Given that B is derived from only 3 raw simulated background events, this number should be read as indicative rather than a robust sensitivity estimate.

Template fit: a binned maximum-likelihood fit of $\mu \times \text{Signal} + \text{Background}$ to the M_{bc} distribution over $[5.20, 5.29]$ GeV (18 bins), with all selection cuts applied except the M_{bc} cut itself, split into simultaneous e and μ channels sharing a common μ , and 1 ab^{-1} of integrated luminosity, gives Table 6.

Table 6: Signal-strength fit results at 1 ab^{-1} .

Channel	μ	Relative precision
Simultaneous $e + \mu$	1.000 ± 0.057	5.7%
e -only	1.000 ± 0.062	6.2%
μ -only	1.000 ± 0.145	14.5%

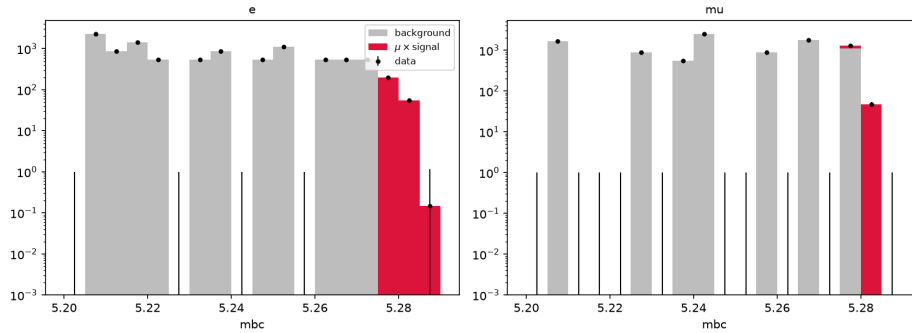


Figure 9: Post-fit M_{bc} distribution from the simultaneous e/μ template fit at 1 ab^{-1} , showing the fitted signal-plus-background templates against the Asimov “data” and the extracted signal strength $\mu = 1.000 \pm 0.057$.

The background-normalization uncertainty in this fit (`bkg_syst`= 0.1) is applied as an independent 10% relative uncertainty *in each bin*, added in quadrature to the bin’s MC-statistical uncertainty, not as a single fully-correlated normalization nuisance parameter across bins; it therefore has some shape-constraining power but does not model a coherent up/down normalization shift of the whole combinatorial template.

Fit validation caveat: $\mu = 1$ is recovered exactly because the fit is performed on the sum of MC templates with no independent Poisson-fluctuated dataset (an Asimov-like fit); this only demonstrates that the fit is unbiased when the data equal the sum of templates, not that its quoted uncertainty has correct coverage. No toy-MC pull study (fitting many independent Poisson-fluctuated pseudo-datasets and checking that the pull distribution of $(\mu_{\text{fit}} - 1)/\sigma_{\text{fit}}$ is a unit Gaussian) was performed in this session. The quoted $\mu = 1.000 \pm 0.057$ should be understood as “the fit returns this statistical uncertainty on an Asimov dataset,” the standard way this framework reports expected sensitivity, but it is not an independently validated coverage guarantee.

Luminosity projection: repeating the same fit with the statistical-only Asimov projection (background shapes treated as known) gives Table 7.

Table 7: Statistics-only Asimov projection of relative precision vs. integrated luminosity.

Integrated luminosity	Expected relative precision
0.1 ab^{-1}	17.4%
0.36 ab^{-1}	9.2%
1 ab^{-1}	5.5%
5 ab^{-1}	2.5%
50 ab^{-1}	0.8%

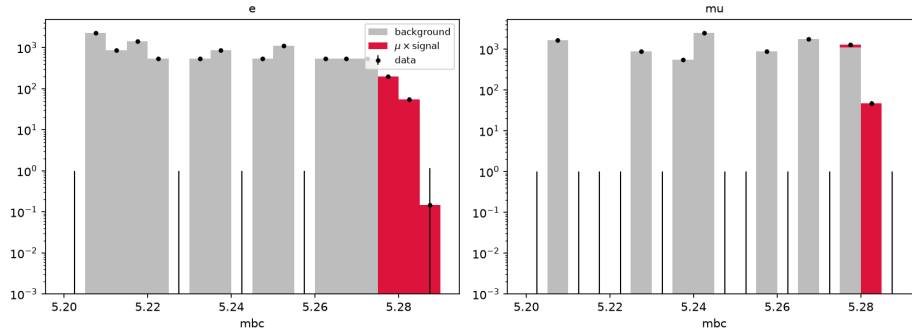


Figure 10: Post-fit M_{bc} distribution for the luminosity-projection configuration at 1 ab^{-1} (statistical-only projection settings).

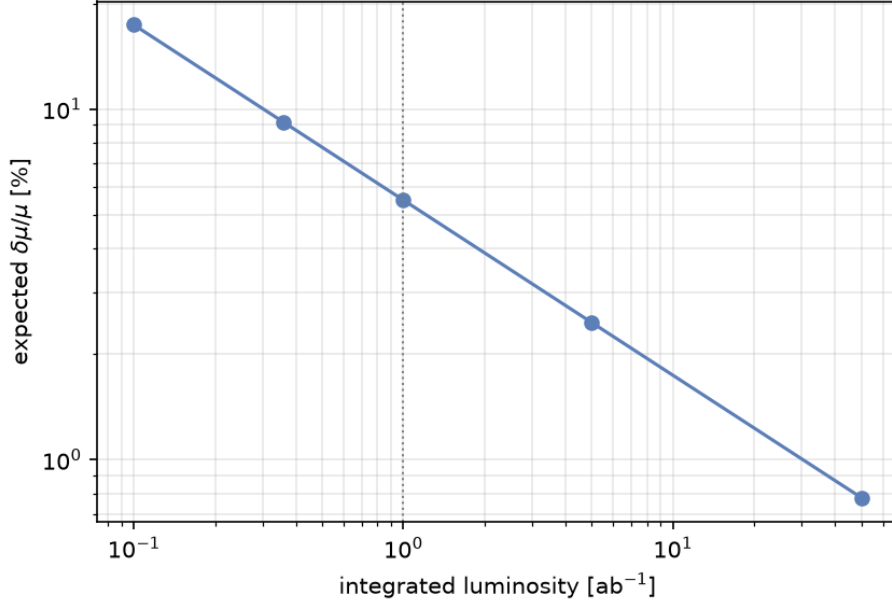


Figure 11: Statistical-only Asimov projection of the expected relative precision on the $B^0 \rightarrow K^{*0} \ell^+ \ell^-$ branching fraction as a function of integrated luminosity, from 0.1 to 50 ab^{-1} .

The 1 ab^{-1} statistical-only projection (5.5%) is close to, but slightly better than, the full fit including the flat 10% per-bin background-normalization systematic (5.7%), showing that at this luminosity the systematic uncertainty budget used here has a small but non-negligible effect; the gap will widen at higher luminosity as the statistical uncertainty continues to shrink while a flat relative systematic does not.

7 Discussion

Limitations of this study, ordered by importance:

1. **Combinatorial MC statistics.** The single largest limitation: the 16×10^6 generated generic- $B\bar{B}$ events yield only 3 raw candidates in the final $M_{\text{bc}} > 5.27$ GeV signal region, and the 6×10^6 continuum events yield 0, driving the wide, asymmetric Poisson interval on the counting background quoted above. The M_{bc} fit mitigates this by using the full $[5.20, 5.29]$ GeV range, but its own combinatorial templates are still built from these same limited samples. A real analysis would use at least an order of magnitude more generic $B\bar{B}$ and continuum MC, or fit an analytic ARGUS + linear shape directly to off-resonance/sideband data as in Wei et al. and the Belle II analysis.
2. **No fit validation beyond the trivial Asimov check.** As stated in Section 6, $\mu = 1$ is guaranteed by construction; no toy-MC pull/bias study was run to validate the coverage of the quoted uncertainty.
3. **No dedicated $\psi(2S)$ MC sample.** The $\psi(2S)$ veto windows are applied identically to the combinatorial and J/ψ samples used here, but no $B \rightarrow \psi(2S) K^{*0}$ MC was generated; the veto's acceptance cost on signal is included, but any residual $\psi(2S)$ leakage into the signal region is not modeled or quantified.

4. **No K/π mis-identification.** Kaon and pion species in the combinatorial samples are assigned from MC truth; real mis-ID would add a further combinatorial background component.
5. **1D M_{bc} fit vs. the published 2D $M_{bc}-\Delta E$ fit.** Using only M_{bc} discards the extra discriminating power of ΔE within the already-applied window and likely makes this study's precision estimate conservative relative to a full 2D fit.
6. **Flat 10% per-bin background-normalization systematic (`bkg_syst`).** This is an unvalidated placeholder; a real analysis would derive the background normalization uncertainty from the $J/\psi K^{*0}$ control region and the M_{bc} sideband fit itself, following the systematics tables in the companion Belle $|V_{cb}|$ analysis [3] and the continuum-suppression and charmonium-veto systematics discussed in BaBar [4].
7. **Seeds not recorded.** The random seeds used to generate the samples in the prior session were not carried forward into this write-up pass; this reproducibility gap should be closed by logging seeds directly into the sample table at generation time in future passes.

8 Conclusion

Following the K^{*0} mass window, R_2 continuum suppression, ΔE and charmonium-veto selection strategy of Wei et al. [Belle] [1] and Belle II [2], the frozen selection for $B^0 \rightarrow K^{*0}(K^+\pi^-)\ell^+\ell^-$ gives a signal efficiency of 34.8% (e) and 31.6% (μ). A cut-and-count analysis in the $M_{bc} > 5.27$ GeV signal region gives $S = 491.7$, $B = 1650$ with a 68% CL Poisson interval of [752, 3255] driven by only 3 raw MC background events. A simultaneous e/μ binned M_{bc} template fit over [5.20, 5.29] GeV recovers the injected signal strength with 5.7% relative precision at 1 ab^{-1} (Asimov check only, not toy-validated), consistent with a statistics-only Asimov projection of 5.5% at the same luminosity, and scaling to 0.8% at 50 ab^{-1} . The $J/\psi K^{*0}$ control channel is a high-purity (90–99%, depending on the treatment of combinatorial MC statistics) sample suitable for future efficiency and PID calibration studies. The dominant limitations are the size of the combinatorial background MC samples, which populate the final signal region with only a handful of raw events, and the absence of a toy-based fit validation; enlarging the MC samples (or replacing the MC combinatorial shape with an analytic ARGUS fit to sideband data, as in the cited analyses) and adding a pull study are the natural next steps.

References

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